

CCR: A Counterfactual Causal Reasoning-Based Method for Cross-View Geo-Localization

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Abstract—Cross-view geo-localization seeks to match geographic locations using images from varied sources, including drones and satellites. Interpreting images captured by drones poses significant challenges due to the varying positions and scales resulting from the camera’s aerial perspective. Traditional approaches have primarily focused on harnessing contextual cues, which may lead to overfitting. Therefore, it is crucial to find an optimal balance between leveraging contextual details and identifying relevant features. To address this, we introduce a novel method for cross-view geo-localization that employs counterfactual causal reasoning (CCR). This method aims to refine the model’s focus, ensuring a balanced emphasis on both the intricate details of the target structure and its broader contextual environment. Our method incorporates an Adaptive Dimension Interaction Block (ADIB), which effectively discerns feature interactions across multiple dimensions, enhanced by counterfactual causal reasoning to improve recognition of target structures and their contexts. In tasks of image-based drone-view target localization and drone navigation, our method achieves superior performance on the University-1652 and SUES-200 benchmark datasets. The code and model files will be made available at <https://github.com/Cyberpunk1998/CCR>.

Index Terms—Image retrieval, drone, cross-view, geo-localization, counterfactual causal reasoning.

I. INTRODUCTION

CROSS-VIEW geo-localization aims to pinpoint geographical coordinates by aligning images from various platforms or vantage points based on their similarity. This domain encompasses a variety of practical applications including drone navigation, event location, precision deliveries, and autopilot systems [1], [2], [3], [4]. Initially, efforts concentrated on synchronizing ground-level images with those from satellites [5], [6], [7], [8]. Given the significant differences in perspectives between ground-level and satellite images, this task presents considerable challenges. However, the field of cross-view geo-localization has experienced substantial advancements due to the rapid development of drone

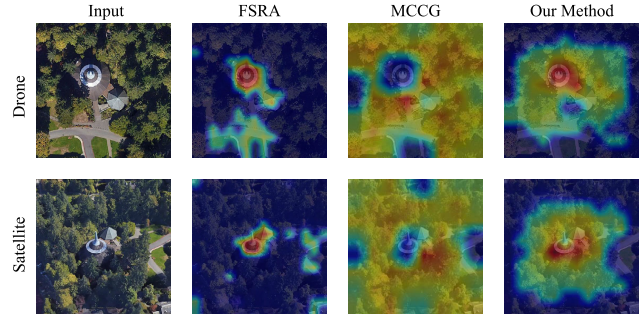


Fig. 1. In our comparative analysis of heatmap generation, we examine the outputs from the FSRA [11], MCCG [12], and our proposed method. The first column depicts the source views from drones and satellites. The second column displays heatmaps derived from the FSRA method, which employs feature segmentation and geographic region alignment. The third column contains heatmaps crafted by the MCCG method, which incorporates a multi-classifier framework. The fourth column demonstrates heatmaps generated by our proposed method. In comparison, our method achieves a more appropriate field of view, enhancing accuracy and detail.

technology [9], [10]. Compared to traditional ground-based geo-localization methods, the integration of aerial imagery from drones has considerably narrowed the gap with satellite images, providing more detailed geographic textures and offering a superior approach for cross-view geo-localization. In the development of this domain, drone-view target localization and drone navigation have emerged as two critical applications. The first application leverages real-time images captured by drones and matched with satellite imagery to facilitate accurate geographic localization through the integration of location information extracted from these satellite images. The second application employs satellite images as query images, matching them in real-time with drone images to enhance drone navigation.

Typically, drone aerial images and satellite images contain abundant contextual information, providing numerous critical clues for cross-view geo-localization. Many existing methods, such as FSRA [11] and MCCG [12], focus on this contextual information. However, an excessive emphasis on a limited aspect of an image’s context may overlook additional locational information and related scene objects. Focusing too intently on contextual details can impede the model’s ability to distinguish between essential and non-essential information, leading to the overlearning of non-essential background details, such as trees and grass. As illustrated in Fig. 1, FSRA overly concentrates on the main body of the building, ignoring the critical locational information provided by the surrounding trees. Conversely, MCCG emphasizes non-essential

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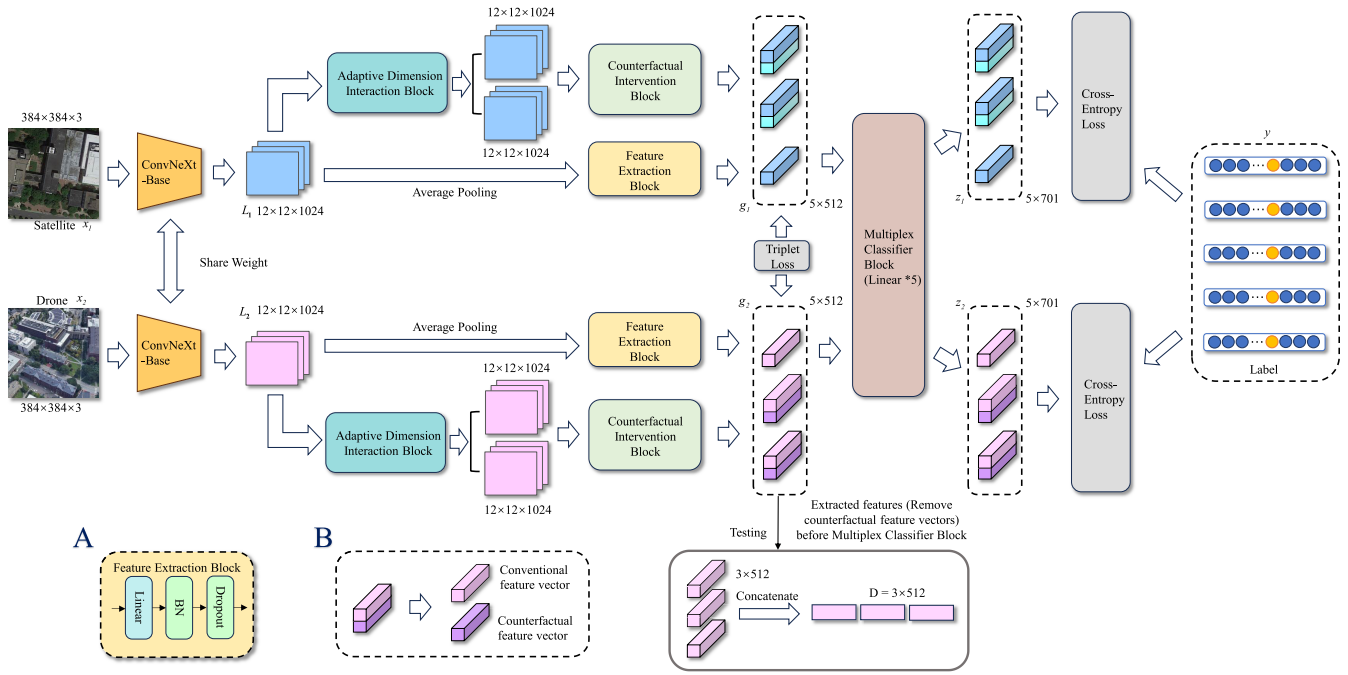


Fig. 2. Network Structure of CCR. The CCR model consists of two primary branches dedicated to processing satellite and drone views, respectively. Due to the similarity in patterns between these views, weight sharing is implemented between the two branches. During training, an input image traverses through the ConvNeXt-Base backbone network to produce feature maps, followed by the adaptive dimension interaction block (ADIB, depicted in Fig. 3) for extracting dimension interaction details. Subsequently, the two sub-feature maps containing extracted dimension interaction information are fed into the counterfactual intervention block (CIB, depicted in Fig. 5) to mitigate excessive attention towards non-essential image features. Lastly, feature vectors extracted by the backbone network and those produced by the CIB block are fed into the multiplex classifier block (MCB) to predict geographic labels. In the testing phase, as counterfactual intervention to suppress attention is unnecessary, the counterfactual features are discarded, resulting in a 3×512 feature representation. The right-angled dashed box A illustrates the structure of the feature extraction block, while box B shows the output vector of the CIB block, comprising conventional feature vectors and counterfactual feature vectors.

contextual details such as the trees, overlooking key features of the central building. Such an excessive focus on non-essential background information can lead to incorrect judgments in environments with homogeneous backgrounds. Prior research [13], [14], [15], [16], [17] has shown that excessive reliance on image contextual information can degrade model performance. Therefore, we advocate for a model that balances attention between the central building and its global contextual surroundings, focusing on essential details while minimizing attention to non-essential elements to maintain an appropriate receptive field.

To address this issue, we draw inspiration from CAL [18] and employ counterfactual causal reasoning to enhance the model's attention learning mechanism. The fundamental concept of counterfactual causal reasoning involves enhancing attention quality by comparing the impacts of factuals (learned attention) and counterfactuals (uncorrected attention) on inference outcomes. By maximizing the difference between factual and counterfactual attention, the model is encouraged to minimize excessive focus on non-essential image features and mitigate overfitting to the global context surrounding the target building. Building on this concept, we introduce a novel method named CCR that implements cross-view matching of satellite and drone images using counterfactual causal reasoning. As illustrated in Fig. 2, CCR comprises two principal branches for processing satellite and drone views, each containing three sub-branches. The first sub-branch employs ConvNeXt-Base [19] as the backbone network for feature map

extraction. The second and third sub-branches extract spatial and channel interaction features using an ADIB block and adjust attention via a counterfactual intervention block (CIB), respectively. Experimental results indicate that our method effectively enhances the model's accuracy.

We summarize our main contributions as follows:

- We proposed CCR for cross-view geo-localization, a novel approach that mitigates model overfitting to non-essential features in images.
- Our method incorporates the ADIB block to extract robust feature representations across various dimensions, utilizing counterfactual causal analysis to process features from both spatial and channel dimensions.
- The implementation of our method has demonstrated unparalleled performance on two significant benchmark datasets, University-1652 [1] and SUES-200 [20], thereby redefining standards in drone-view target localization and drone navigation.

The structure of this paper is methodically organized as follows: Section II provides a concise overview of relevant prior research; Section III details the CCR approach for cross-view geo-localization that we propose; Section IV presents the experimental results of our research; Section V concludes the paper with a summary discussion of our findings.

II. RELATED WORKS

In this section, we succinctly review prior studies relevant to our methodology, specifically focusing on the fields of

cross-view geo-localization and the application of causal reasoning in computer vision.

A. Cross-View Geo-Localization

Research in cross-view geo-localization has garnered considerable attention due to its broad applicability across various domains, primarily focusing on two key sub-tasks: matching ground to satellite images and drone to satellite images. Within the realm of aligning ground to satellite images, seminal studies such as DeepGeo [5] and UrbanGeo [8] have addressed challenges in urban environments, providing extensive datasets to correlate street-level photographs with angled aerial images. CVUSA [21] and CVACT [22] have contributed valuable resources for scholarly research by compiling comprehensive sets of panoramic street scenes and satellite images. Moreover, VO [23] has focused on identifying image scenes rather than camera positions, utilizing over a million ground-to-air image pairs. The recent introduction of the VIGOR [24] dataset, which includes challenges such as high-rise building obstructions and shadows, has added complexity to the field. In the sub-task of drone to satellite image matching, the prevalent dataset is University-1652 [1], which includes 1,652 buildings from 72 universities worldwide and pioneers tasks in drone-view target localization and drone navigation, thereby enhancing the model's ability to generalize viewpoint-invariant features. The newly introduced SUES-200 [20] dataset, comprising images collected at four different altitudes from 200 locations, introduces fresh perspectives and challenges for cross-view research.

As research into cross-view geo-localization deepens, scholars have recognized the significant impact of loss functions on model performance. Loss functions are essential metrics for quantifying the discrepancies between a model's predictions and the desired outcomes. Selecting an appropriate loss function is crucial for developing efficient models. In the domain of image retrieval, the adoption of cross-entropy loss [25], triplet loss [26], [27], and contrastive loss [28] has become standard, consistently yielding superior results. Pioneering work in cross-view tasks by Zheng et al. [29] has integrated instance and verification loss, leading to substantial advancements. Additional contributions by Hu et al. [30] include the introduction of a weighted soft-margin ranking loss, which has enhanced the precision of retrievals and accelerated the training phase. In improving cross-view geo-localization accuracy, Luo et al. [22] developed the BNNeck framework by merging ID loss with triplet loss. Concurrently, Sun et al. [31] formulated circle loss, integrating ID loss and triplet loss into a unified strategy to optimize pair similarity.

The essence of cross-view geo-localization is spatial alignment and feature matching to ensure consistency across perspectives. Initial research focused on recognizing images that are closely related to geographic locations through hand-designed features [32], [33], [34], [35], [36]. With advances in deep learning, neural networks have shown strong feature extraction capabilities. Workman and Jacobs [37] used a pre-trained convolutional neural network (CNN) to extract features from ground and aerial images. In their

subsequent work [21], they fine-tuned this network using a contrastive loss function to bring similar sample pairs closer together. However, ground images, rich in in-depth information, cannot be fully characterized by neural networks alone. Consequently, some researchers employ multi-scale feature aggregation to capture image features across different scales. For example, Li et al. [38] used multi-scale attention to reduce distortion from polar coordinate transformations in ground images, while Royston et al. [39] improved ground view images with semantic-driven data augmentation and used a multi-scale attention network for feature aggregation. Additionally, other researchers have used semantic-guided segmentation to help models capture fine details in images. For instance, Li et al. [40] guided feature extraction and representation with fine-grained location supervision; Xue et al. [41] improved feature reliability and distinctiveness by using implicit semantic cues for feature detection and description.

To address the matching issue between drone and satellite view images, Ding et al. [42] introduced the LCM method for cross-view matching, which reconceptualizes the image retrieval challenge as a classification task using ResNet [43] as the backbone for feature extraction. Wang et al. [44] developed the LPN model for cross-view geo-localization, which employs a square-ring partition strategy in feature segmentation to enhance the integration of contextual information. Dai et al. [11] utilized a Vision Transformer (ViT) [45] as the structural basis of their model, segmenting feature maps into distinct regions guided by the generated heatmaps to align various feature areas across disparate views. Shen et al. [12] proposed MCCG, a multiple-classifier cross-view geo-localization method based on ConvNeXt [19], to get a full representation of features. Building on the MCCG concept, we proposed the CCR method, which aims to address the issue of over-reliance on contextual information in existing cross-view geo-localization research.

B. Causal Reasoning in Computer Vision

The essence of causal reasoning involves identifying potential causal relationships through the observation and analysis of data. By eliminating confounding factors, causal reasoning focuses on elucidating true causal relationships rather than merely superficial correlations, thereby enhancing model performance and effectiveness across various scenarios. With rapid advancements in deep learning technology, researchers are integrating causal reasoning with deep learning to address challenges in areas such as fairness [46], interpretable machine learning [47], reinforcement learning [48], adversarial learning [49], and natural language processing [50]. As a potent method, causal reasoning mitigates dataset biases and extends its applications to visual tasks. For instance, Lv et al. [51] utilized causal intervention modules, causal decomposition modules, and adversarial mask modules to achieve more robust representation learning. Huang et al. [52] introduced causal relationships in object detection tasks, treating context as a confounding factor. Zhang et al. [53] constructed a structured causal model involving pixels, context, and labels, which eliminates the influence of contextual information on class

activation map generation. Wang et al. [54] proposed a causal attention module that identifies contextual confounding factors in an unsupervised manner. Rao et al. [18] employed counterfactual causal relationships to refine attention mechanisms, encouraging models to focus on broader, useful information. Liu et al. [55] introduced a causality-inspired representation consistency (CRC) framework that learns causal variables from normal videos to detect abnormal events. Moreover, to tackle multimodal utterance-level label bias and word-level context bias in multimodal emotion classification tasks, Yang et al. [56] proposed a multimodal counterfactual reasoning emotion analysis framework based on causal relationships. By comparing factual and counterfactual results, this framework makes unbiased decisions from biased observations. Consequently, to ensure an appropriate receptive field for processing drone and satellite views, we apply counterfactual causal reasoning to the cross-view geo-localization task. This approach aims to mitigate the model's overfitting to the global context around the target and reduce the impact of non-essential contextual information, thereby enhancing the model's matching accuracy.

III. PROPOSED METHOD

This section provides a detailed examination of the components of the CCR method. Fig. 2 presents the complete architectural representation of the method. Initially, Section III-A outlines the problem formulation for cross-view geo-localization between drone and satellite views. Subsequently, Section III-B details the architecture employed for feature extraction in the model. Section III-C then discusses the Counterfactual Intervention Block (CIB), which aims to reduce the model's reliance on non-essential image features through counterfactual causal reasoning. Finally, Section III-D explicates the model training objectives.

A. Problem Formulation

Consider a cross-view geo-localization dataset denoted as $\{x_j, y\}$, where x_j represents the input images and y denotes their corresponding labels. To distinguish images from different vantage points, we use subscript j , where $j = 1, 2$, representing satellite (x_1) and drone (x_2) view images, respectively. The label y belongs to the set $1, \dots, R$, where R indicates the number of locations in the dataset, such as $R = 701$ for University-1652 and $R = 80$ for SUES-200. Here, L_j denotes the feature map generated by the backbone network, g_j represents the feature vector associated with the images, and z_j corresponds to the prediction vector for the input images.

B. Feature Extraction Architecture

1) *Backbone Feature Extraction*: As illustrated in Fig. 2, the proposed CCR model comprises two primary branches dedicated to processing satellite and drone views. Due to the aerial nature and similar patterns observed in both views, weight sharing is implemented between these branches. The feature maps for each primary branch are extracted using the

pre-trained ConvNeXt-Base network [19]. ConvNeXt represents an advancement over ResNet [43], integrating design elements from Swin Transformer [57] and achieving performance comparable to Vision Transformer [45], [57] networks. ConvNeXt-Base, a variant of ConvNeXt, shows a notable increase in floating-point operations per second (FLOPs) when processing images at a resolution of 384×384 pixels, in contrast to Swin Transformer. Consequently, input images are resized to 384×384 pixels to ensure compatibility with the ConvNeXt-Base layer $\mathcal{F}_{\text{ConvNeXt-Base}}$, which produces feature maps L_j with dimensions $B \times H \times W \times C$. In this context, B represents the batch size, H indicates the height, W signifies the width, and C denotes the channel count of the feature map. The procedure for feature extraction via the backbone network is formalized as follows:

$$L_j = \mathcal{F}_{\text{ConvNeXt-Base}}(x_j) \quad (1)$$

After generating feature maps L_j , a global average pooling (GAP) process condenses these maps into the vector p_j of dimension $B \times 1 \times C$. When processing a single image, setting $B = 1$ simplifies the output to a vector of dimensions $1 \times 1 \times C$. The process is described as follows:

$$p_j = \text{Avepool}(L_j) \quad (2)$$

The vector p_j is then processed through the feature extraction block (FEB), denoted as $\mathcal{F}_{\text{FEB}}$, to yield refined feature vectors $g_{j1} \in \mathbb{R}^{1 \times 1 \times 512}$ for the network. The structure of the feature extraction block (FEB), $\mathcal{F}_{\text{FEB}}$, is depicted in the dashed rectangle A at the bottom left of Fig. 2, consisting of three components: a linear layer, a batch normalization layer, and a dropout layer. The extraction process within the FEB block proceeds as follows:

$$g_{j1} = \mathcal{F}_{\text{FEB}}(p_j) \quad (3)$$

2) *ADIB Block Feature Extraction*: Typically, basic features from images but often fail to fully interpret or utilize contextual information. This limitation becomes particularly evident in satellite and drone imagery, which is rich in contextual details crucial for discerning location-specific features. To address this issue, we have introduced an ADIB block following the feature map extraction by the backbone network. This block incorporates an enhanced triple attention mechanism [58], as illustrated in Fig. 3. By establishing dependencies among different dimensions, this module effectively encodes channel and spatial information without reducing dimensionality, thereby preserving more original information and enhancing feature interaction.

In the original triple attention mechanism, while parallel processing of average and max pooling preserves the tensor's rich representation, max pooling tends to overly emphasize the maximum values in local regions, potentially neglecting key secondary features. To address this issue, we replaced max pooling with standard pooling to ensure that important features are not overlooked. Additionally, given the varying impacts of different pooling methods on image feature extraction, assigning equal weights to these methods is suboptimal. Consequently, we introduced two adaptive weights to more effectively fuse features extracted by average pooling and

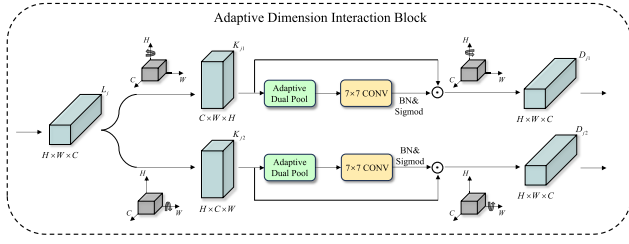


Fig. 3. Structure of the ADIB block. The ADIB block consists of two distinct sub-branches: the upper sub-branch encodes interactions between the width W and channel C dimensions, while the lower sub-branch captures interactions between the height H and channel C dimensions. Both sub-branches employ a rotational mechanism to enhance connectivity among dimensions. Each sub-branch undergoes a series of operations: adaptive dual pooling, convolution, batch normalization, and Sigmoid activation, culminating in broadcasting for element-wise multiplication. This process generates feature maps D_{ji} , which are congruent in size with the feature map L_j extracted by the backbone network. The adaptive dual pooling mechanism is further illustrated in Fig. 4.

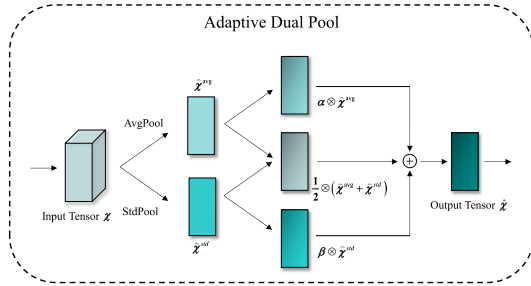


Fig. 4. Schematic of adaptive dual pooling. Its function is to adaptively aggregate average pooling and standard pooling.

standard pooling. We refer to this strategy as adaptive dual pooling, anticipating that it will improve the model's adaptability in complex environments. The schematic of adaptive dual pooling, illustrated in Fig. 4, processes an input tensor χ through both average pooling and standard pooling to produce tensors $\hat{\chi}^{\text{avg}}$ and $\hat{\chi}^{\text{std}}$, respectively. These tensors are subsequently combined via an adaptive mechanism, producing the output tensor $\hat{\chi}$. This process is represented as follows:

$$\hat{\chi}^{\text{avg}} = \text{AvgPool}_d(\chi) \quad (4)$$

$$\hat{\chi}^{\text{std}} = \text{StdPool}_d(\chi) \quad (5)$$

$$\hat{\chi} = \frac{1}{2} \otimes (\hat{\chi}^{\text{avg}} \oplus \hat{\chi}^{\text{std}}) \oplus \alpha \otimes \hat{\chi}^{\text{avg}} \oplus \beta \otimes \hat{\chi}^{\text{std}} \quad (6)$$

In the equation above, $\alpha, \beta \in [0, 1]$, are two trainable floating parameters optimized using Stochastic Gradient Descent (SGD). The variable d specifies the dimension across which average pooling and standard pooling operations are conducted.

As depicted in Fig. 3, the backbone network feature map L_j , having dimensions $H \times W \times C$, is input into two sub-branches of the ADIB block, $\mathcal{F}_{\text{ADIB}}$, for the extraction of interactive features within the spatial and channel domains. Each sub-branch processes the feature map L_j through a rotational transformation, producing K_{j1} and K_{j2} , with dimensions $C \times W \times H$ and $H \times C \times W$, respectively. After this transformation, an adaptive dual pooling process reshapes K_{j1} and K_{j2} into dimensions of $C \times W \times 1$ and $H \times W \times 1$, respectively. Following this, attention weights are computed via a convolution layer, which is succeeded by batch normalization and Sigmoid

activation. These weights are then broadcast for element-wise multiplication with K_{j1} and K_{j2} , respectively. Ultimately, the feature maps are reorganized to their original dimensions of $H \times W \times C$. The complete procedure is outlined below:

$$[D_{j1}, D_{j2}] = \mathcal{F}_{\text{ADIB}}(L_j) \quad (7)$$

In the equation presented above, D_{j1} and D_{j2} are the two sub-feature maps generated by the ADIB block, which encapsulate the results of interactive feature extraction from the spatial and channel domains, respectively. The incorporation of the ADIB block design throughout the model significantly enhances its adaptability and expressive capabilities, facilitating the network's ability to emphasize key features in images.

Subsequently, the sub-feature maps, D_{j1} and D_{j2} , generated by the ADIB block, are processed by the CIB block. This block generates both conventional and counterfactual feature vectors. The detailed implementation of this process is discussed in Section III-C.

C. The Counterfactual Intervention Block (CIB)

In current research, most attention modules utilize a weakly supervised approach to learn visual attention, characterized by reliance on loss functions for guidance. This approach lacks strong supervisory signals that could explicitly direct the training process. Properly leveraging the causal relationship between prediction and attention could guide the model to effectively distinguish between essential and non-essential features in images, thus mitigating biases towards prejudiced information during region learning. Consequently, inspired by CAL [18], we introduce a CIB block following the ADIB block. This block specifically suppresses the model's focus on non-essential features. The detailed implementation process is described below:

1) *Counterfactual Causal Reasoning*: Before introducing the CIB block, we delineate the process of counterfactual causal reasoning using causal graph language. As depicted in the lower right corner of Fig. 5, D represents the feature map input to the CIB block, A is the learned attention map, and $G(A)$ represents the features extracted based on this attention map. The link $D \rightarrow A$ denotes the production of the conventional attention map by the Counterfactual Generation Block (CGB) using the input feature map. Additionally, the link $(D, A) \rightarrow G(A)$ indicates a joint determination of the feature outcome, influenced by both the feature map and the attention map. In the revised causal graph, we sever the link $D \rightarrow \bar{A}$, signifying that the counterfactual attention map $\bar{A}$ is not derived from the input feature map D . Consequently, the features $G(\bar{A})$ extracted from the counterfactual attention map $\bar{A}$ are deemed erroneous. According to the diagram, g^{effect} is calculated by subtracting $G(\bar{A})$ from $G(A)$. Essentially, the efficacy of the attention can be assessed by the degree to which it refines the final feature extraction outputs compared to those derived from inaccurate attention.

2) *The CGB Block*: As illustrated in Fig. 5, upon receiving the sub-feature map D_{ji} , the CIB block processes it through the CGB block $\mathcal{F}_{\text{CGB}}$ to generate the conventional attention map A_{ji} and the attention map $\bar{A}_{ji}$. Fig. 6 depicts the network

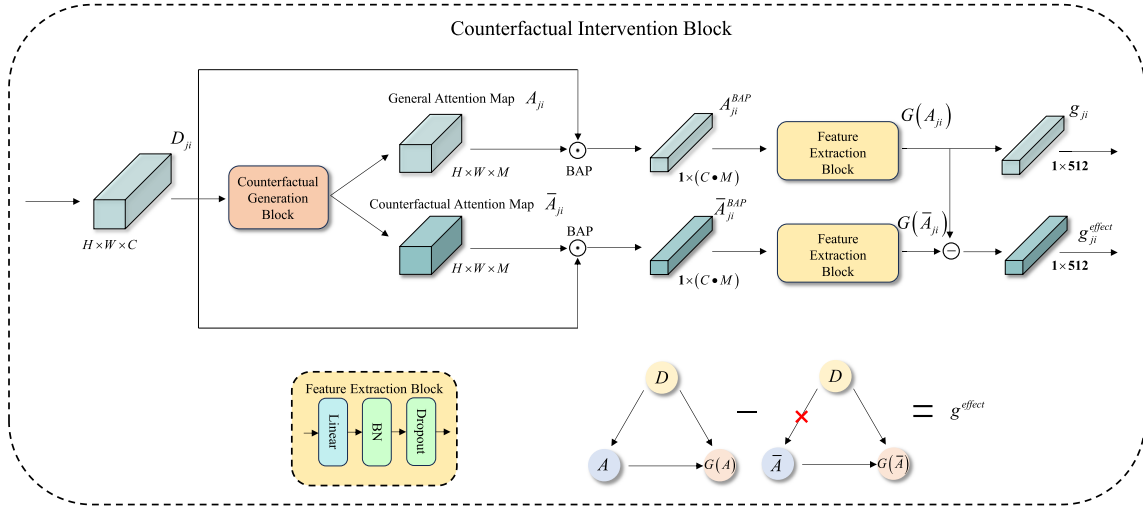


Fig. 5. Network structure of the CIB block. The feature vector D_{ji} is processed by the counterfactual generation block (CGB), which generates both the conventional attention map A_{ji} and the counterfactual attention map $\bar{A}_{ji}$. The network structure of the CGB block is depicted in Fig. 6. These attention maps are utilized to apply soft weighting to the feature map, generating A_{ji}^{BAP} and $\bar{A}_{ji}^{BAP}$. Subsequently, the FEB block extracts features, and the result of counterfactual feature extraction is subtracted from conventional feature extraction, yielding both conventional and counterfactual feature vectors.

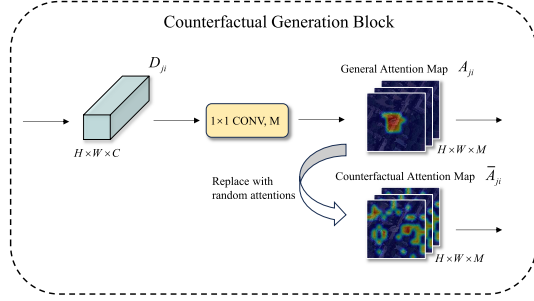


Fig. 6. Network structure of the counterfactual generation block (CGB). The counterfactual attention map $\bar{A}_{ji}$ is produced by substituting the conventional attention map A_{ji} with randomly generated attention.

structure of the CGB block. The sub-feature map D_{ji} first passes through a convolution layer with an output channel of M and a 1×1 kernel, followed by a ReLU activation layer, producing a conventional attention map A_{ji} of shape $H \times W \times M$. Subsequently, by substituting A_{ji} with random attention, the counterfactual attention map $\bar{A}_{ji}$ is generated. This process is represented as follows:

$$[A_{ji}, \bar{A}_{ji}] = \mathcal{F}_{CGB}(D_{ji}) \quad (8)$$

3) *Counterfactual Intervention Process*: As previously described, the CGB block converts the sub-feature map D_{ji} into the conventional attention map A_{ji} and the counterfactual attention map $\bar{A}_{ji}$, both with dimensions $H \times W \times M$. Subsequently, these two attention maps undergo element-wise multiplication with the sub-feature map to softly weight the feature map. Following this, they are aggregated via a global average pooling operation, producing A_{ji}^{BAP} and $\bar{A}_{ji}^{BAP}$, both sized $1 \times 1 \times (C * M)$. This operation is collectively referred to as bilinear attention pooling, denoted $\mathcal{F}_{BAP}$. The process is outlined as follows:

$$\begin{aligned} A_{ji}^{BAP} &= L_2(\mathcal{F}_{BAP}(D_{ji}, A_{ji})) \\ &= L_2(\text{GAP}(D_{ji} \odot A_{ji})) \end{aligned} \quad (9)$$

$$\begin{aligned} \bar{A}_{ji}^{BAP} &= L_2(\mathcal{F}_{BAP}(D_{ji}, \bar{A}_{ji})) \\ &= L_2(\text{GAP}(D_{ji} \odot \bar{A}_{ji})) \end{aligned} \quad (10)$$

In the equation above, $\odot$ denotes element-wise multiplication, and L_2 signifies L2 normalization.

After obtaining A_{ji}^{BAP} and $\bar{A}_{ji}^{BAP}$, they are processed through the FEB block $\mathcal{F}_{FEB}$ to extract features, resulting in $G(A_{ji})$ and $G(\bar{A}_{ji})$. Unlike the FEB block described in Section III-B, this feature block incorporates different linear layer dimensions. Subsequently, subtracting $G(\bar{A}_{ji})$ from $G(A_{ji})$ generates the counterfactual feature vector $g_{ji}^{effect} \in \mathbb{R}^{1 \times 1 \times 512}$, while $G(A_{ji})$ serves as the conventional feature vector $g_{ji} \in \mathbb{R}^{1 \times 1 \times 512}$. The process is outlined as follows:

$$g_{ji} = G(A_{ji}) = \mathcal{F}_{FEB}(A_{ji}^{BAP}) \quad (11)$$

$$\begin{aligned} g_{ji}^{effect} &= G(A_{ji}) - G(\bar{A}_{ji}) \\ &= \mathcal{F}_{FEB}(A_{ji}^{BAP}) - \mathcal{F}_{FEB}(\bar{A}_{ji}^{BAP}) \end{aligned} \quad (12)$$

Consequently, we obtain a quintet of feature vectors: g_{j1} , g_{j2} , g_{j3} , g_{j4} and g_{j5} (where $g_{j4} = g_{j2}^{effect}$ and $g_{j5} = g_{j3}^{effect}$ defined). To minimize feature disparity across identical targets captured from varied viewpoints, triplet loss is applied to the quintet of feature vectors. The detailed methodology for this implementation is described in Section III-D.

D. Training Objectives

In the context of cross-view matching, where identical targets viewed from multiple perspectives are grouped into a unified category, the task is transformed into a classification challenge. The cross-entropy loss function is widely utilized to quantify the divergence between predicted probabilities and actual class labels, owing to its high precision in measurement. By applying the cross-entropy loss function to the quintet of feature vectors outlined in Section III-C, we enhance the

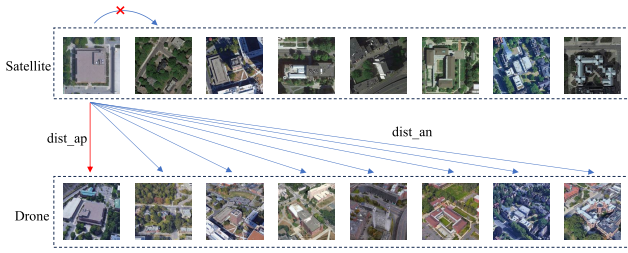


Fig. 7. Schematic representation of triplet loss calculation. The left column shows 8 images from the satellite view, whereas the right column shows 8 images from the drone view. The term ' $dist_{ap}$ ' denotes the distance metric for pairs within the same view, while ' $dist_{an}$ ' quantifies the distance between pairs from different views. The red $\times$ shows that images with the same view don't compute triplet loss.

accuracy of our classification. The process is described in detail as follows:

$$z_{ji} = \mathcal{F}_{M-Classifier}(g_{ji}) \quad (13)$$

$$\hat{p}(y | x_{ji}) = \frac{\exp(z_{ji}(y))}{\sum_{c=1}^C \exp(z_{ji}(c))} \quad (14)$$

$$CL_{ji} = -\log(\hat{p}(y | x_{ji})) \quad (15)$$

In this formulation, $\mathcal{F}_{M-Classifier}$ denotes the multiplex classifier block (MCB), $z_{ji}(y)$ represents the logarithm of the actual geographic label $\hat{p}(y | x_{ji})$ is the predicted probability that the geographic label y is correct.

Triplet loss is utilized to refine the feature space, thereby reducing the distance between features of identical targets captured from different viewpoints. Typically, feature distance is measured using either Euclidean distance or cosine similarity. In accordance with prior research [11], [12], we employ Euclidean distance to measure these distances. The triplet loss function is defined as follows:

$$d(a, x) = \|a - x\|_2 \quad (16)$$

$$TL = \max(\|d(a, p) - d(a, n) + M\|, 0) \quad (17)$$

Here, $\|\cdot\|_2$ denotes the L2 norm operation, where a represents the anchor sample, p and n represent the positive and negative samples, respectively. The term M represents the margin value, which is uniformly set to 0.3 across all experiments. In cross-view matching tasks, the fundamental goal is to match images from different views, rather than those within the same view. Consequently, triplet loss is applied exclusively to images from differing views, as illustrated in Fig. 7.

The total loss function comprises both cross-entropy loss and triplet loss, with the detailed computational formula provided below:

$$L_{total} = \frac{1}{3} \sum_{i=1}^3 \sum_{j=1}^2 (CL_{ji} + TL_{ji}) \quad (18)$$

wherein, i denotes the i^{th} feature vector among the quintet of feature vectors outlined in Section III-C. Here, $j = \{1, 2\}$ corresponds to images from different views. The entire model is optimized using L_{total} .

IV. EXPERIMENT

In Section IV-A, descriptions of two public benchmark datasets for cross-view geo-localization are provided. Section IV-B elaborates on the implementation details used in the experiments. Section IV-C presents a comparison of the results obtained from our method against those from existing methods. Section IV-D reports on the experimental results of the ablation study.

A. Datasets and Evaluation Protocols

This work addresses the correspondence challenge between satellite and drone perspectives, with a particular focus on drone-view target localization and drone navigation tasks. To thoroughly assess the approach introduced in this paper, two large-scale geo-localization datasets, University-1652 and SUES-200, were utilized. Table I and Table II provide detailed statistics on the number of images from various perspectives in these datasets, respectively.

1) *University-1652*: The University-1652 dataset is a comprehensive resource designed to address challenges in cross-view geo-localization. This large-scale dataset amalgamates imagery from drone, satellite, and ground levels, featuring contributions from 1652 distinct locations associated with 72 universities worldwide. Each site is represented by a collection of 54 drone-captured images taken from various heights and angles, a single satellite image, and multiple street-level images. The drone imagery is generated via a simulation interface from Google Earth, whereas the satellite and ground-level images are sourced from Google Maps and additional search engine resources. University-1652 is distinguished as the inaugural dataset tailored for geo-localization from a drone's perspective, supporting two essential cross-view matching tasks: drone-view target localization (Drone to Satellite) and drone navigation (Satellite to Drone). Fig. 8 presents illustrative examples from the University-1652 dataset, showcasing a broad spectrum of viewing angles.

Table I provides a detailed configuration of the University-1652 dataset, delineating separate training and testing sets encompassing 701 and 951 locations from 33 and 39 universities, respectively, with no shared images between the dataset parts. The training subset comprises 37,854 drone-view images, 701 satellite images, and 11,604 street-level images. Within the scope of drone-view target localization (Drone to Satellite), the query subset consists of 37,855 drone-view images, while the gallery subset comprises 951 satellite-view images, including 250 images used as distractors. In the context of drone navigation (Satellite to Drone), the query subset includes 701 satellite-view images, while the gallery subset comprises 51,355 drone-view images, with 13,500 images designated as distractors.

2) *SUES-200*: The SUES-200 dataset, recently introduced, is specifically designed for multi-altitude, multi-scene cross-view geo-localization and captures authentic drone imagery at altitudes of 150 m, 200 m, 250 m, and 300 m. This dataset enhances the exploration of scene diversity, a challenge not fully addressed in the University-1652 dataset, by including

TABLE I

STATISTICAL INFORMATION FROM THE UNIVERSITY-1652 DATASET, INCLUDING COUNTS OF DRONE, SATELLITE, AND STREET VIEW IMAGES, CLASSIFICATIONS OF BUILDINGS, AND THE ENUMERATION OF UNIVERSITIES. THESE ARE DIVIDED INTO TRAINING AND TESTING SETS, WITH A STRICT NO-OVERLAP POLICY ENFORCED

Division	Views	Images	Classes	Universities
Training Set	Drone View	37,854	701	33
	Satellite View	701	701	
	Street View	11,640	701	
Testing Set	Drone Query	37,855	701	39
	Satellite Query	701	701	
	Street Query	2,579	701	
	Drone Gallery	51,355	951	
	Satellite Gallery	951	951	
	Street Gallery	2,921	793	

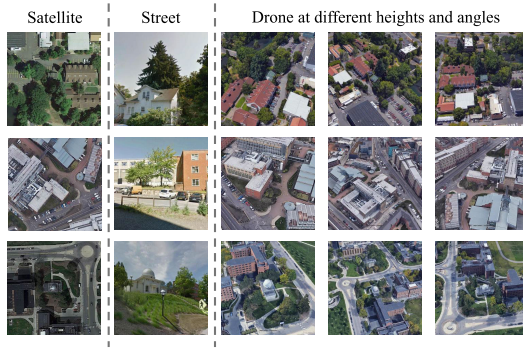


Fig. 8. Some images from the University-1652 dataset with different viewpoints.

TABLE II

THE SUES-200 DATASET CONTAINS STATISTICAL INFORMATION, INCLUDING THE NUMBER OF IMAGES FOR EACH LOCATION, THE TOTAL COUNT OF LOCATIONS, AND A COMPREHENSIVE TALLY OF IMAGES FROM EACH VIEW WITHIN THE TRAINING AND TESTING SETS

Division	Views	Images	Locations	Total
Training Set	Drone View	6,000	120	24,000
	Satellite View	120	120	120
Testing Set	Drone Query	4,000	80	16,000
	Satellite Query	80	80	80
	Drone Gallery	10,000	200	40,000
	Satellite Gallery	200	200	200

varied environments such as parks, educational institutions, lakes, and urban infrastructure. It comprises 200 unique locations, with 120 allocated to the training set and 80 designated for testing. Each location is represented by one satellite view image and a corresponding set of drone view images distributed across four altitude tiers, with each tier containing 50 images. Fig. 9 displays illustrations from the SUES-200 dataset, while Table II offers a detailed statistical breakdown.

3) *Evaluation Protocols*: Within this experimental framework, Recall@K (R@K) and Average Precision (AP) were

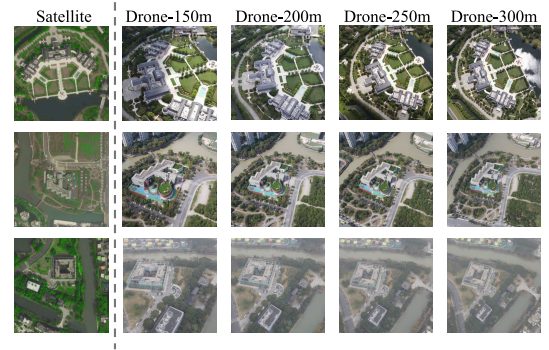


Fig. 9. Some images from the SUES-200 dataset with different viewpoints.

employed as metrics to evaluate the model's performance. R@K measures the proportion of relevant outcomes accurately identified within the top K retrieved results. A higher R@K value indicates stronger retrieval capability of the model. Conversely, the AP metric evaluates the ranking accuracy when all relevant results are accurately retrieved. It calculates the average precision at the point of retrieval for each relevant result. An increase in AP value signifies an improved performance of the model in prioritizing relevant results.

B. Implementation Details

In this study, ConvNext-Base, pretrained on ImageNet, was employed as the backbone network for the CCR method. The CCR architecture integrates an ADIB block to capture feature information across multiple dimensions, complemented by a CIB block designed to enhance perception of the target building and its contextual environment. For model parameter initialization, the classifier component utilizes the Kaiming initialization strategy [59]. Regarding image processing, all images from the University-1652 dataset were resized to 384×384 pixels prior to model input. For the SUES-200 dataset, the input image size was maintained at 512×512 pixels. Additionally, to improve the model's generalization capabilities, image augmentation techniques, including random cropping, padding, and flipping, were uniformly applied to all input images.

During the training phase, SGD was employed as the optimization technique, with a momentum of 0.9, a weight decay of 0.0005, and a batch size of 8. The initial learning rates were set at 0.003 for the backbone layers and 0.01 for other layers, with a systematic tenfold decrease scheduled at epochs 80 and 120, within a total of 200 epochs. The architecture was developed using Pytorch, and computations were accelerated by an Nvidia RTX 4090 GPU with 24GB of memory.

During the assessment phase, the model's performance was evaluated by computing cosine similarity distances to compare features between the query image and each potential match in the gallery.

C. Comparison With Other Methods

1) *Results of CCR on University-1652*: Table III presents the outcomes of the CCR method on the University-1652 dataset, achieving an R@1 of 92.54% and an AP of 93.78% for

TABLE III

COMPARATIVE RESULTS OF CCR AND STATE-OF-THE-ART METHODS ON THE UNIVERSITY-1652 DATASET, WITH OPTIMAL OUTCOMES EMPHASIZED IN BOLDFACE

Method	Year	Drone to Satellite		Satellite to Drone	
		R@1	AP	R@1	AP
LCM [42]	2020	66.65	70.82	79.89	65.38
LPN [44]	2021	75.93	79.14	86.45	74.79
PCL [10]	2021	83.27	87.32	91.78	82.18
FSRA [11]	2022	82.25	84.82	87.87	81.53
PAAN [60]	2022	84.51	86.78	91.01	82.28
MBF [61]	2023	89.05	90.61	93.15	88.17
MCCG [12]	2023	89.64	91.32	94.30	89.39
TransFG [62]	2024	84.01	86.31	90.16	84.61
SeGCN [63]	2024	89.18	90.89	94.29	89.65
CCR (Ours)	\	92.54	93.78	95.15	91.80

drone-view target localization (Drone to Satellite). For drone navigation (Satellite to Drone), the method recorded an R@1 of 95.15% and an AP of 91.80%. These results represent a 2% enhancement in performance relative to the benchmark previously established by the state-of-the-art MCCG method.

2) *Results of CCR on SUES-200*: Table IV demonstrates that the CCR method achieves significant results in drone-view target localization (Drone to Satellite), with R@1 rates of 87.08%, 93.57%, 95.42%, and 96.82%, and AP rates of 89.55%, 94.90%, 96.28%, and 97.39% across four different altitudes. Similarly, for drone navigation (Satellite to Drone), the CCR method reports R@1 rates of 92.50%, 97.50%, 97.50%, and 97.50%, and AP rates of 88.54%, 95.22%, 97.10%, and 97.49% at these altitudes. Except for the 150m altitude metrics for R@1 and AP, and the R@1 at 300m altitude in drone navigation, the CCR method maintains state-of-the-art standings in all other categories, demonstrating robust and consistent performance across various altitudes.

D. Ablation Studies

To assess the impact of each component in the proposed CCR method, a series of ablation studies were conducted on the University-1652 dataset.

1) *Effect of the ADIB Block and the CIB Block*: Table V details the results of ablation studies conducted on the CCR method. The introduced ADIB and CIB blocks both significantly enhanced the model's performance. Relative to the baseline, incorporating these blocks led to a 6% increase in R@1 and a 5% improvement in AP for the task of drone-view target localization (Drone to Satellite), and a 2% rise in R@1 and a 6% increase in AP for drone navigation (Satellite to Drone). Among these enhancements, the ADIB block was instrumental in driving performance gains, while the CIB block also made significant contributions to the observed improvements.

To gain deeper insights into the effects of different pooling strategies within the ADIB block, we evaluated the outcomes of three distinct approaches: Average Pooling (AvgPool), Max Pooling (MaxPool), and Standard Pooling (StdPool), along with their synergistic combinations. Empirical evidence,

detailed in Table VI, indicates that the combination of Average and Standard Pooling provides a competitive advantage in performance. This advantage is intuitively understood as resulting from the synergistic integration of information extracted through Average Pooling with that from Standard Pooling.

2) Effect of the Backbone Network in Cross-View:

To evaluate the impact of various backbone networks on model performance, a series of tests were conducted on the University-1652 dataset using an array of backbones: ResNet-50, ResNet-101, ConvNeXt-Tiny, ConvNeXt-Small, and ConvNeXt-Base. Empirical results, detailed in Table VII, demonstrate the marked superiority of ConvNeXt-Base over ResNet-50 in cross-view geo-localization tasks, with significant improvements in both R@1 and AP metrics, exceeding 20%. Compared to ConvNeXt-Tiny, ConvNeXt-Base provided an incremental advantage, improving these metrics by approximately 2%.

3) *Effect of Input Image Size*: To achieve an optimal balance between model efficacy and computational efficiency, we investigated the impact of varying input image dimensions on model performance. This study evaluates the trade-offs involving image resolution—where reducing image size could obscure critical details—and computational demands, as larger images increase complexity and memory requirements. Our findings, outlined in Table VIII, show a consistent improvement in model efficacy with incremental increases in input image resolution, especially notable when increasing from 224 pixels to 384 pixels. However, further enlargement to 512 pixels led to a slight decline in performance. Through this ablation study, we aim to provide guidance on selecting the appropriate input image size tailored to specific hardware resource conditions.

4) *Effect of the CGB Block Output Channel M* : The number of output channels (M) in the CIB block plays a crucial role in the architectural design of the model. A smaller M might result in attention maps that fail to capture essential details, while an excessively large M increases computational complexity and memory usage. Therefore, selecting an appropriate M requires careful consideration. Empirical evidence presented in Table IX demonstrates that the model achieves optimal performance in drone-view target localization (Drone to Satellite) with M set at 32. For drone navigation (Satellite to Drone), peak performance is reached when M is set at 128. To strike a balance between precision and computational efficiency, we have selected a configuration of M at 32.

5) Effect of the FEB Block's Feature Vector Dimensions:

To examine the impact of output feature vector dimensions on the feature extraction phase, we adjusted the parameter settings of the FEB block, thereby altering the size of its resultant feature vector. Evidence presented in Table X shows that CCR's optimal performance was achieved with the feature vector dimension set at 512.

6) *Robustness Validation of CCR*: To rigorously investigate the robustness of the CCR model, we designed and implemented seven distinct testing methods. Initially, the model's resilience to image position transformations was evaluated using three methods: Left Black Pad (LBP), Left Flip Pad

TABLE IV
COMPARATIVE ANALYSIS OF CCR AND STATE-OF-THE-ART METHODS ON THE SUES-200 DATASET,
WITH OPTIMAL OUTCOMES EMPHASIZED IN BOLDFACE

Drone to Satellite								
Method	150 m		200 m		250 m		300 m	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
SUES-200 Baseline [20]	55.65	61.92	66.78	71.55	72.00	76.43	74.05	78.26
LCM [42]	43.42	49.65	49.42	55.91	54.47	60.31	60.43	65.78
LPN [44]	61.58	67.23	70.85	75.96	80.38	83.80	81.47	84.53
FSRA [11]	59.18	65.28	74.88	79.20	82.67	85.76	88.88	90.82
MBF [61]	85.62	88.21	87.43	90.02	90.65	92.53	92.12	93.63
MCCG [12]	82.22	85.47	89.38	91.41	93.82	95.04	95.07	96.20
SeGCN [63]	90.80	92.32	91.93	93.41	92.53	93.90	93.33	94.61
CCR (Ours)	87.08	89.55	93.57	94.90	95.42	96.28	96.82	97.39

Satellite to Drone								
Method	150 m		200 m		250 m		300 m	
	R@1	AP	R@1	AP	R@1	AP	R@1	AP
SUES-200 Baseline [20]	75.00	55.46	85.00	66.05	86.25	69.94	88.75	74.46
LCM [42]	57.50	38.11	68.75	49.19	72.50	47.94	75.00	59.36
LPN [44]	83.75	66.78	88.75	75.01	92.50	81.34	92.50	85.72
FSRA [11]	73.75	63.70	86.25	78.02	91.25	84.83	93.25	89.88
MBF [61]	88.75	84.74	91.25	89.95	93.75	90.65	96.25	91.60
MCCG [12]	93.75	89.72	93.75	92.21	96.25	96.14	98.75	96.64
SeGCN [63]	93.75	92.45	95.00	93.65	96.25	94.39	97.50	94.55
CCR (Ours)	92.50	88.54	97.50	95.22	97.50	97.10	97.50	97.49

TABLE V
ABLATION EXPERIMENT RESULTS OF THE PROPOSED CCR METHOD

Method	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
w/o ADIB & CIB	86.56	88.66	93.44	85.89
w/o CIB	91.76	93.13	94.86	91.53
CCR	92.54	93.78	95.15	91.80

TABLE VI
ABLATION EXPERIMENT RESULTS OF DIFFERENT POOLING STRATEGIES
OF THE ADIB BLOCK. IN THE TABLE, THE ‘+’ SIGN REPRESENTS THE
COMBINATION OF DIFFERENT POOLING STRATEGIES

Pooling Strategy	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
AvgPool	91.50	92.87	94.72	90.86
MaxPool	92.04	93.41	95.44	91.16
StdPool	92.36	93.56	95.72	91.48
AvgPool + MaxPool	91.46	92.86	95.15	91.37
AvgPool + StdPool	92.54	93.78	95.15	91.80
MaxPool + StdPool	92.41	93.58	95.58	91.57
AvgPool + MaxPool + StdPool	91.54	92.93	95.15	91.49

(LFP), and Top-Left Black Pad (T-LBP). As depicted in Fig. 10, LBP involves modifying the target building’s position by adding a black padding of width P to the left side and cropping an equivalent width from the right side. LFP applies a mirrored padding of width P to the left side of the image while simultaneously cropping from the right side. T-LBP introduces a black padding similar to LBP at the top of the

TABLE VII
ABLATION EXPERIMENT RESULTS OF DIFFERENT BACKBONE NETWORKS

Backbone Network	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
ResNet-50	70.92	74.78	85.16	71.28
ResNet-101	74.78	78.29	85.88	74.67
ConvNeXt-Tiny	90.60	92.07	93.30	89.95
ConvNeXt-Small	91.69	93.07	94.58	91.28
ConvNeXt-Base	92.54	93.78	95.15	91.80

TABLE VIII
ABLATION EXPERIMENT RESULTS OF DIFFERENT IMAGE SIZES

Image Size	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
224	90.13	91.73	94.15	89.78
256	90.39	91.97	94.72	90.57
320	91.07	92.50	94.29	90.46
384	92.54	93.78	95.15	91.80
512	91.31	92.70	94.58	90.60

image and crops from the bottom. Additionally, to evaluate the model’s performance under conditions of restricted viewing angles, we utilized the All Black Pad (ABP) method, which adds black paddings of width P around the image to simulate limited visibility due to obstacles, urban density, or specific camera angles. As demonstrated in Table XI, setting the padding width to 60 results in the AP of LBP, LFP, and T-LBP decreasing by 4.12%, 9.86%, and 14.69%, respectively,

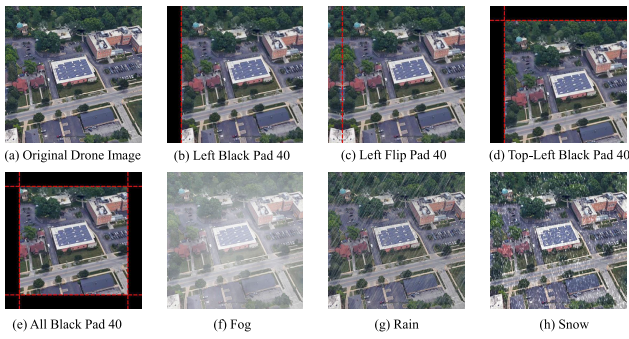


Fig. 10. Examples of BlackPad, FlipPad, and synthesized data under extreme weather conditions. The red dashed line indicates the boundary of the padding.

TABLE IX
ABLATION EXPERIMENT RESULTS OF THE CIB BLOCK'S
OUTPUT CHANNEL COUNT M

M Size	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
16	91.77	93.08	93.72	91.11
32	92.54	93.78	95.15	91.80
64	92.27	93.55	95.29	91.37
128	92.06	93.33	95.58	91.89
256	90.82	92.33	94.01	90.57

TABLE X
ABLATION EXPERIMENT RESULTS OF THE FEB BLOCK'S
FEATURE VECTOR DIMENSIONS

Feature Vector Dimensions	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
256	91.71	93.04	95.15	91.31
512	92.54	93.78	95.15	91.80
768	91.89	93.18	95.01	91.31

highlighting the model's maintained high recognition accuracy despite significant positional transformations. In comparison, LFP significantly decreases accuracy due to edge information confusion, while T-LBP results in a more substantial loss of accuracy due to bidirectional positional shifts. Furthermore, setting ABP to 60 led to only a 2.58% decrease in AP, underscoring the CCR model's robustness in scenarios of limited visibility.

Furthermore, to assess the model's performance under extreme weather conditions, we simulated three weather environments—fog, rain, and snow—and incorporated these scenarios into drone imagery, as illustrated in Fig. 10. The data presented in Table XII elucidates the impact of extreme weather conditions on drone-based target localization and navigation tasks. These findings indicate that rainy conditions most significantly impair model performance, followed by snowy conditions, while foggy conditions exert the least impact. Notably, in drone navigation tasks, the Recall@1 metric in foggy conditions surpasses that in normal environments. We postulate that this phenomenon arises because fog tends to reduce the high-frequency components of images, diminish background noise, and thereby enhance the visibility of target features. In summary, despite the challenges posed by extreme

TABLE XI
THE AP ACCURACY VALUES AND DECLINE RATES OF THE PROPOSED
METHOD CCR UNDER DIFFERENT PAD SIZES FOR LBP, LFP,
T-LBP, AND ABP

Pad Pixel	LBP AP%	LFP AP%	T-LBP AP%	ABP AP%
0	93.78-0	93.78-0	93.78-0	93.78-0
10	93.70-0.08	93.51-0.27	93.69-0.09	93.58-0.20
20	93.49-0.29	92.70-1.08	93.07-0.71	93.42-0.36
30	92.96-0.82	91.53-2.25	91.66-2.12	93.02-0.76
40	92.21-1.57	89.61-4.17	89.25-4.53	92.59-0.79
50	90.98-2.80	87.28-6.50	84.68-9.10	91.97-1.81
60	89.66-4.12	84.10-9.68	79.09-14.69	91.20-2.58

TABLE XII
RESULTS OF ROBUSTNESS STUDIES UNDER
DIFFERENT WEATHER EXTREMES

Weather Conditions	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
Normal	92.54	93.78	95.15	91.80
Fog	91.65	93.03	95.72	91.16
Rain	87.19	89.19	94.44	87.24
Snow	89.83	91.44	95.15	89.68

TABLE XIII
ABLATION EXPERIMENT RESULTS OF THE DRONE-VIEW
IMAGE'S NUMBER

Drone-view image's number	Drone to Satellite		Satellite to Drone	
	R@1	AP	R@1	AP
1	85.07	87.36	93.30	84.50
3	89.52	91.19	94.72	88.77
9	91.63	93.00	94.72	90.76
18	91.35	92.77	94.86	90.94
27	92.10	93.45	95.01	91.04
54	92.54	93.78	95.15	91.80

weather, the model demonstrates high recognition accuracy, highlighting the robustness of the CCR model under such adverse conditions.

7) *Effect of Drone-View Image Quantity*: A single drone-view image may not capture the diverse visual cues of a target effectively, which could lead to imprecise location estimates. To address this issue, the University-1652 dataset includes an extensive collection of 54 drone-view images per target location, taken from various heights and angles. The integration of multiple drone-view images substantially improves model performance. To evaluate the compatibility with satellite-view images, experiments were conducted using varying subsets of 1, 3, 9, 18, and 27 images per target location. The data in Table XIII reveal a clear positive correlation between the number of drone-captured images and the accuracy of the model, suggesting that a larger collection of such images significantly enhances precision.

E. Visualization Of Qualitative Result

As illustrated in Fig. 11, we visualized the results of two tasks on the CCR model using the University-1652 dataset.

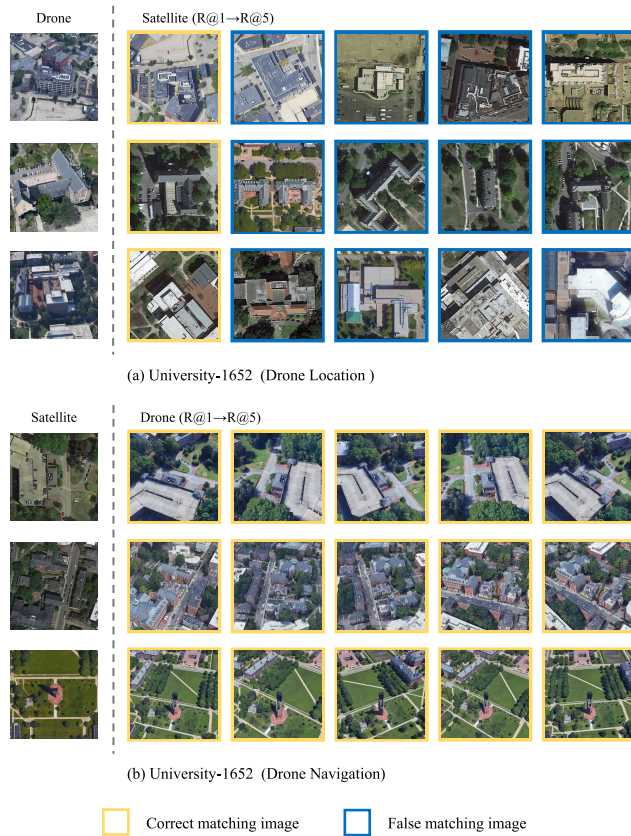


Fig. 11. Qualitative image retrieval results. (a) displays the top five retrieval outcomes for the drone-view target localization task using the CCR method within the University-1652 dataset. (b) displays the top five retrieval outcomes for the drone navigation task using the CCR method within the University-1652 dataset. Here, yellow boxes indicate correct matches, and blue boxes represent incorrect matches.

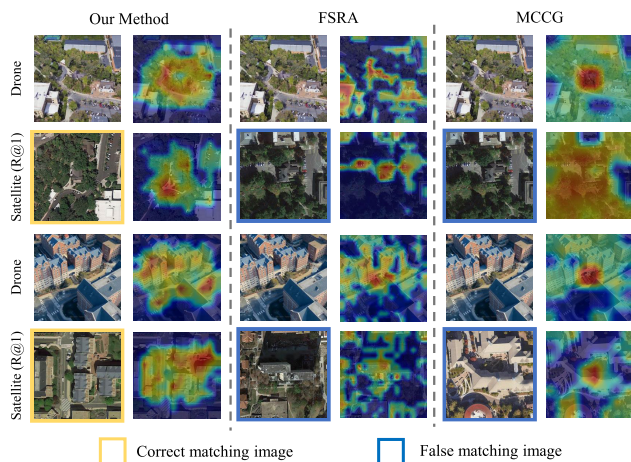


Fig. 12. CCR, FSRA, MCCG in the University-1652's drone-view target localization task's Top-1 retrieval results comparison. The left column shows the retrieval results of the proposed CCR, the middle column for FSRA, and the right column for MCCG. Among them, the yellow box indicates the correct matching results, and the blue box indicates the incorrect matching results.

We observed that the CCR model is capable of accurately retrieving results in many similar scenes. In the drone view target localization task, we randomly selected three drone view images from the query set and displayed the top five matching results, all of which were correctly retrieved by

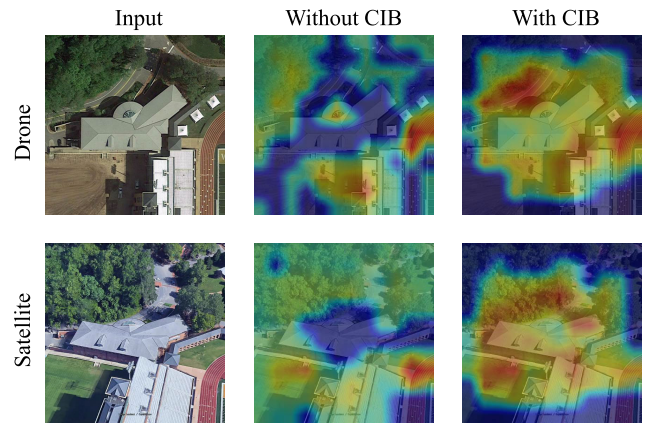


Fig. 13. Comparison of model heatmaps with and without the CIB block. The leftmost images show the input drone and satellite views. The middle column displays the heatmaps of the model without the CIB block. The rightmost column presents the heatmaps of the model with the CIB block.

the CCR model. Similarly, in the drone navigation task, after randomly selecting three satellite view images, the top five matching results also demonstrated completely accurate retrieval performance.

To demonstrate the effectiveness of the CCR model, we showcase two examples in Fig. 12 comparing the Top-1 retrieval results of the CCR model against the baseline models FSRA and MCCG in drone-view target localization. The results indicate that the CCR model consistently retrieved the correct satellite views, whereas FSRA and MCCG did not. Attention heatmap analysis revealed that the CCR model demonstrated a higher level of focus on processing critical information compared to FSRA and MCCG, effectively minimizing the impact of misleading and distracting information. This finding confirms that the CCR model possesses superior target recognition and localization capabilities, achieved by mitigating overfitting to the global context.

Furthermore, Fig. 13 provides a comparison of heatmaps generated by the CCR model, with and without the inclusion of the CIB block. The heatmap that incorporates the CIB block exhibits a more rational receptive field. It effectively reduces undue emphasis on non-target elements such as trees, ground, and roads, while enhancing focus on target buildings. This indicates that the inclusion of counterfactual causal reasoning within the CIB block significantly improves the model's ability to perceive target buildings and their contextual surroundings, thereby enhancing retrieval accuracy.

V. CONCLUSION

Contemporary cross-view geo-localization approaches often over-rely on contextual features, which can lead to reduced model performance. To address this issue, we introduce a cross-view geo-localization method that incorporates counterfactual causal reasoning, called CCR. This method aims to reduce the dependence on non-critical information in the global context and increase the focus on critical information, such as target buildings. Enhanced by the ADIB block, the CCR method adeptly extracts detailed and distinctive feature representations from cross-view images, facilitating both

spatial and channel interactions. The effectiveness of CCR is demonstrated through its exceptional performance on the large benchmark datasets University-1652 and SUES-200, where it surpasses current state-of-the-art methods. Notably, the CCR method exhibits robust performance. In future research, we plan to extend counterfactual causal reasoning to a broader range of cross-view matching tasks, including the alignment of ground panoramic images with satellite views.

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